📦 SUPPLY CHAIN ANALYTICS REPORT

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1. PROJECT OVERVIEW

This project performs an end-to-end exploratory data analysis (EDA) on a supply chain dataset to understand operational efficiency, delivery performance, pricing behaviour, customer satisfaction, and return patterns.

The goal is to convert raw transactional data into actionable business insights using Python-based data analysis techniques (Pandas, NumPy, Matplotlib, Seaborn), without applying machine learning models.

The analysis focuses on identifying inefficiencies in order fulfilment, delivery performance, supplier and warehouse operations, product-level behaviour, and customer feedback patterns.

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2. BUSINESS PROBLEM STATEMENT

Modern supply chain systems often suffer from inefficiencies that directly impact cost, delivery speed, and customer satisfaction.

This analysis aims to answer:

- How efficient is the delivery process across orders?
- Which factors contribute to delays or cancellations?
- How do pricing, quantity, and discounts affect order behavior?
- What patterns exist in returns and customer ratings?
- Are there operational inconsistencies across suppliers and warehouses?

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3. DATASET OVERVIEW

The dataset contains **12,000 records** and **16 features** representing supply chain transactions.

Key Columns:

- order_id, product, quantity
- city, supplier, warehouse

- price, transport_cost, discount
- order_date, delivery_date
- delivery_status, returned, customer_rating
- payment_method, notes

Data Characteristics:

- Mixed data types (numerical + categorical)
- Missing values in customer_rating
- Inconsistent entries such as "Unknown"
- Unstructured text in notes column

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4. DATA CLEANING SUMMARY

The following preprocessing steps were performed:

- Missing values identified in customer_rating (~17%)
- Converted order_date and delivery_date into datetime format
- Standardized categorical variables (supplier, warehouse, city)
- Handled inconsistent values such as "Unknown"
- Ensured logical consistency of dataset

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5. EXPLORATORY DATA ANALYSIS (EDA)

5.1 Product & Order Analysis

- Price ranges from ₹1001 to ₹49,995
- Quantity ranges from 1 to 19 units per order

Insight:
There is high variability in product pricing and order quantity, indicating diverse demand behaviour across products.

5.2 City & Warehouse Analysis

- Orders span across Bangalore, Hyderabad, Chennai, Mumbai
- Presence of "Unknown" warehouse values

Insight:
Warehouse mapping is incomplete in some cases, indicating operational tracking gaps.

5.3 Delivery Performance Analysis

- Delivery status includes: Delivered, Cancelled, Pending

Insight:
Supply chain performance is multi-stage, not purely success/failure, indicating operational complexity in fulfilment.

5.4 Cost & Discount Analysis

- Transport cost is relatively stable (mostly 200–300 range)
- Discounts range from 0% to 15%

Insight:
Logistics cost is standardized, while pricing strategy uses controlled discounting.

5.5 Customer Rating Analysis

- Average rating is approximately 3.0 / 5
- Missing values present in ratings (~17%)

Insight:
Customer satisfaction is moderate, and feedback collection is incomplete.

5.6 Return Analysis

- Returned field indicates product/order failure cases

Insight:
Returns act as a strong indicator of operational inefficiency or product dissatisfaction.

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6. KEY BUSINESS INSIGHTS

- Delivery performance is inconsistent across operations
- Product demand and pricing show high variability
- Warehouse mapping inconsistencies exist in dataset
- Customer satisfaction is moderate with missing feedback data
- Returns are a strong signal of supply chain inefficiency
- Operational behaviour is multi-factor driven, not dependent on a single variable

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7. KPI FRAMEWORK

OTIF (On-Time In-Full Delivery)

Measures delivery reliability:

OTIF = Delivered Orders / Total Orders

Fill Rate

Measures fulfilment efficiency:

Fill Rate = (Delivered + Pending Orders) / Total Orders

Return Rate

Measures product or service failure:

Return Rate = Returned Orders / Total Orders

Revenue per Order

Measures value contribution:

Revenue = (Price × Quantity) - Discount

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8. BUSINESS RECOMMENDATIONS

- Improve warehouse mapping accuracy by eliminating "Unknown" entries
- Optimize delivery operations to reduce inconsistencies
- Improve customer feedback collection systems
- Reduce return rates through better quality control
- Implement KPI monitoring (OTIF, Fill Rate, Return Rate)
- Align inventory and supply decisions with demand behaviour

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9. CONCLUSION

This project demonstrates how raw supply chain data can be transformed into meaningful business insights using exploratory data analysis.

The findings highlight inefficiencies in delivery systems, inconsistencies in warehouse tracking, moderate customer satisfaction levels, and variability in product performance.

These insights can help organizations improve operational efficiency, reduce cost leakage, and enhance customer experience.

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10. FUTURE SCOPE

- Power BI dashboard development for interactive insights
- Predictive modelling for delivery delays
- Automated KPI tracking system
- Supplier performance benchmarking